

SATYAM SAKRAL

Delhi, India • +91 9958503242 • satyamsakral@gmail.com • [Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Accomplished Software & AI/ML Engineer with expertise in backend system architecture, computer vision engineering, and LLM post-training pipelines. Proven track record of designing scalable Django and FastAPI web services, re-engineering SQL database queries for low-latency performance, and building real-time WebRTC streaming applications. Adept at training PyTorch convolutional neural networks, engineering multi-modal dataset pipelines, and conducting RLHF alignment for production language models.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

AI / ML & LLMs: Retrieval-Augmented Generation (RAG), Fine-Tuning, Prompt Engineering, LangChain, PyTorch, OpenCV, Scikit-learn, LLM Evaluation & Alignment

Web & Backend: FastAPI, Django, Spring Boot, RESTful APIs, WebRTC, Socket.io

Databases: PostgreSQL, Supabase (RLS), MySQL, MongoDB

Tools & Workflows: Git, GitHub, Docker, Postman, Semantic Search, Vector Embeddings

WORK EXPERIENCE

Ethara AI — LLM Trainer Intern

Apr 2026 – Jun 2026

- **LLM Post-Training & Optimization:** Spearheaded fine-tuning, RLHF alignment, and evaluation phases for large language models, significantly elevating contextual coherence and task execution accuracy.
- **Dataset Engineering & Quality Control:** Engineered data pipelines to curate, filter, and deduplicate multi-modal datasets, reducing model hallucination rates and eliminating training biases.
- **Flux AI Core Integration:** Collaborated with senior research engineers to construct evaluation benchmarks and reinforcement feedback loops for core Flux AI framework pipelines.

Doosra College — Django Developer Intern

Jul 2023 – Sep 2023

- **Full-Stack Inventory Backend:** Architected and deployed a centralized Inventory Management System using Django REST framework and SQL to streamline logistics tracking across departmental modules.
- **Database Optimization:** Re-engineered complex SQL queries and indexing strategies, achieving a **60% reduction** in data retrieval latency for high-throughput database operations.
- **Analytics Dashboard Integration:** Collaborated with cross-functional teams to integrate interactive data visualization dashboards into the operational pipeline.

PROJECTS

Study AI – Adaptive Learning & Neural RAG Platform | [GitHub](#)

Python | FastAPI | LangChain | Supabase

- **Zero-Trust BYOK Architecture:** Built a multi-provider AI Gateway supporting OpenAI, Anthropic, Gemini, Groq, and OpenRouter with AES-GCM (Fernet) encryption for client API key storage.
- **Multi-Source Neural RAG Pipeline:** Implemented multi-format document parsing (PDF, DOCX, PPTX, YouTube Transcripts) coupled with vector semantic search for context-aware document Q&A.
- **Adaptive Planner & Dynamic Assessments:** Designed automated study plan scheduling algorithms based on topic difficulty and learning speed, paired with real-time exam feedback generation.

FaceMask Detection – Real-Time Computer Vision Classifier | [GitHub](#)

Python | PyTorch | OpenCV

- **CNN Classifier Architecture:** Trained a deep Convolutional Neural Network achieving **~81% accuracy** in detecting facial mask compliance via live video feeds.
- **Data Augmentation & Pipeline:** Built automated image pre-processing and data augmentation pipelines to withstand ambient lighting variations and occlusion.

VividChat – WebRTC Real-Time Video & Messaging System | [GitHub](#)

Django | WebRTC | Socket.io

- **Peer-to-Peer Streaming Architecture:** Developed a low-latency video streaming web app using Django backend controllers and WebRTC peer connection protocol.
- **Scalable Signaling Infrastructure:** Integrated Socket.io for dynamic room management, group calling capabilities, and real-time text broadcasting.

EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU)

Delhi, India

- **Master of Computer Applications (MCA)** – JIMS Rohini | CGPA: **8.2 / 10** | *Expected: Jun 2026*
- **Bachelor of Computer Applications (BCA)** – SGTBIMIT | CGPA: **7.7 / 10** | *Graduated: Jun 2024*

CERTIFICATIONS

- **Specialized Program in AI & ML with Drone Tech** – TiHAN, IIT Hyderabad
- **Java Programming Masterclass** – Udemy Certification